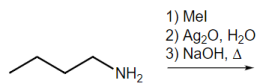
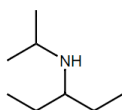


The final exam will include all the chapters we learned. The new chapter (CH23, which wasn't a part of any midterms) will be included as well, and the following pages have the practice problems for this chapter.

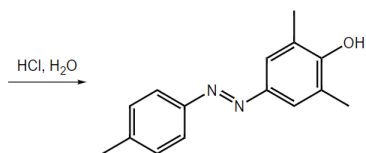
For the previous chapters, please use the three midterm exams as well as their practice problems (you should be able to find answers in Moodle, either the pdf files or whiteboard images). The final exam questions for those chapters will be similar or identical to those previous questions.

Predict starting materials or products

1) HCl, H₂O
2) NaBH₃CN, H₂O



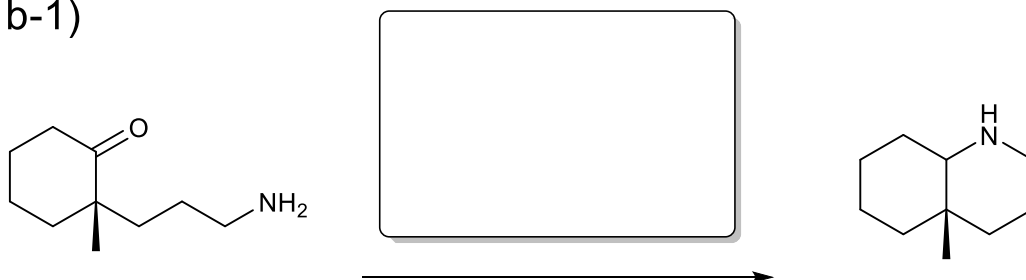
1) MeI
2) Ag₂O, H₂O
3) NaOH, Δ



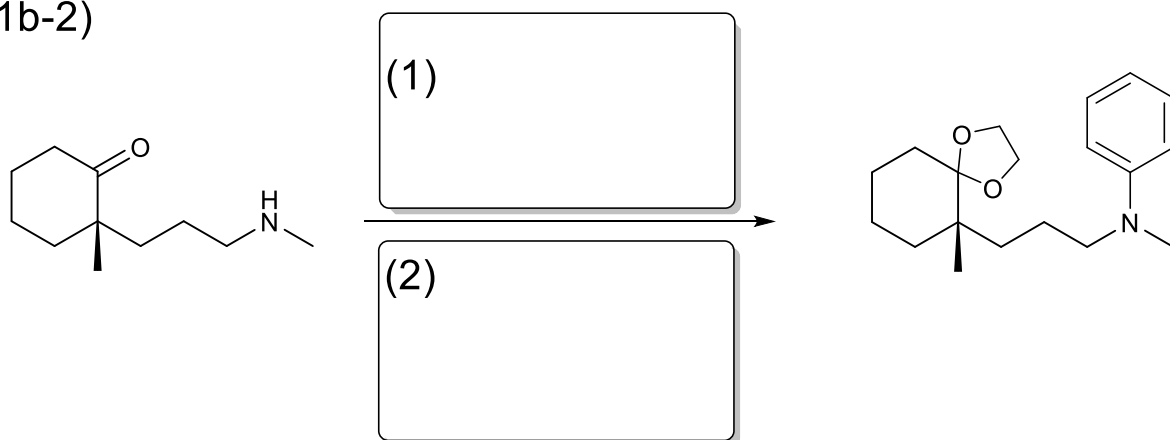
HCl, H₂O

1b. Provide the reagents for the following reactions:

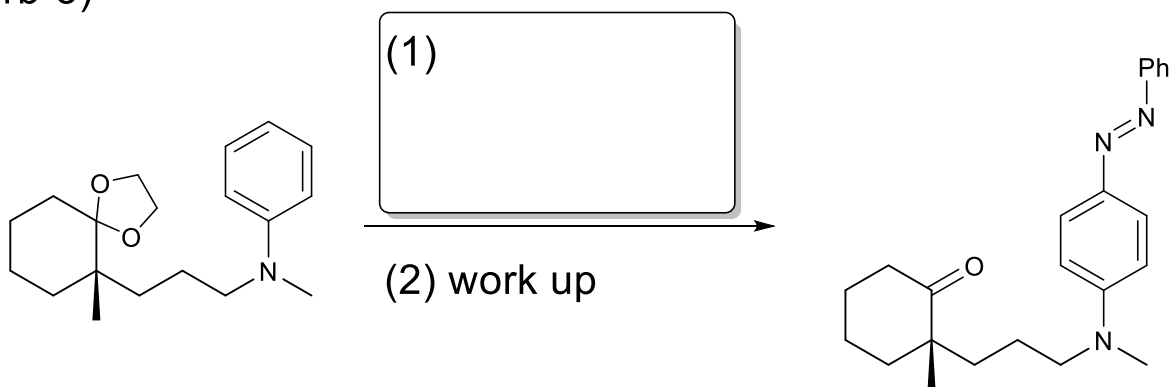
1b-1)



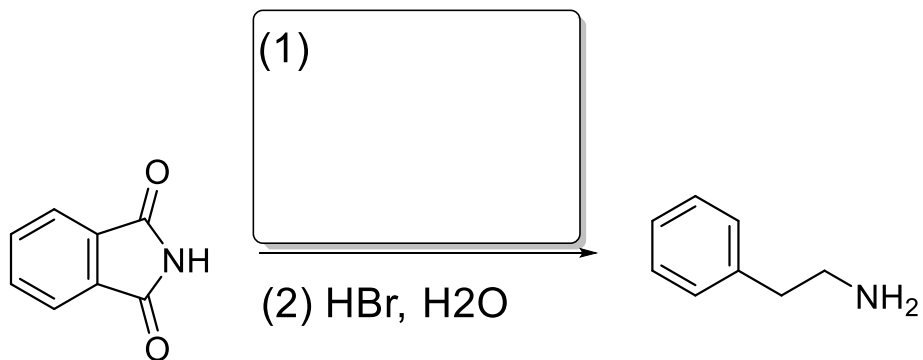
1b-2)



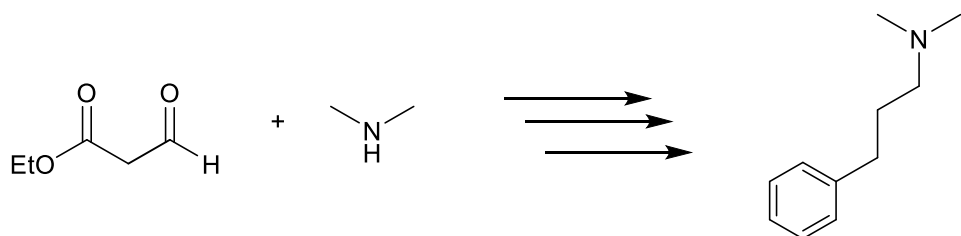
1b-3)



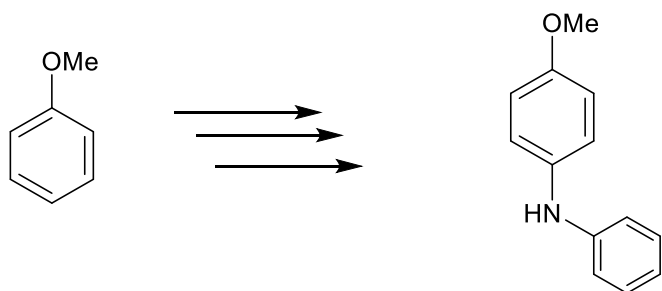
1b-4)



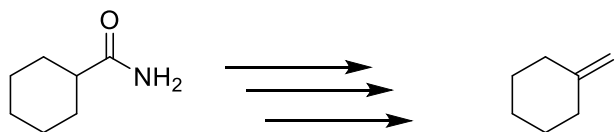
Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right one. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.



Design a synthetic route of the following molecule on the left to the right ones. There are multiple approaches, but a typical design would require ~4 steps. Provide reaction conditions and product structure of each step clearly.

